

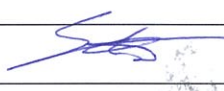
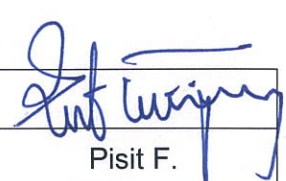


PTT Global Chemical Public Company Limited
GCEP-PP-005

PAINTING

Rev. 1

1 January 2021

				
1	1-Jan-21	Vasathon A.	Sittichai W.	Pisit F.
0	17-Feb-16	Kasem S.	Weerachat I.	Somsuk K.
Revision	Issued Date	Prepare	Check	Approve

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SUMMARY OF DOCUMENT REVISIONS				
Rev. No.	Date Revised	Section Revised	Revision Description	Initiator
0	17-Feb-16	All	First Issue	
1	1-Jan-21	4.1	Add painting requirement of austenitic stainless steel, Fiber reinforced plastic (FRP) and Glass reinforced epoxy (GRE).	Vasathon A.
		4.4	Revise content	
		5.1.1, 5.1.2, 5.1.3	Add content for paint system proposal and approval	
		5.1.9	Add content for criteria of paint system selection	
		5.3	Revise content	
		5.4.4	Add requirement for painting of dissimilar weld joint	
		6.6	Add requirement of stainless steel surface preparation	
		6.7	Add requirement of surface preparation by power tool	
		6.10, 6.12, 6.13	Revise content for surface preparation inspection requirement	
		8.1	Allow to use copper slag as abrasive material	
		10.1.5	Add content	
		12.1	Revise content for Inspector's qualification requirement	
		12.4	Revise acceptant criteria for DFT	
		16.1	Revise letter sizing	
		Table 1.1, 1.2, 1.3	Delete existing paint system table and replace with new one, paint trade name and brand was removed	

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This work was done by a working group of GC Static Equipment Community of Practice (SE CoP) , Review - initial approved by Static Equipment Discipline Technical Committee(SE DTC) , and Endorsed by GC GCEP committee.

At the time this revision was completed, the SE CoP AUTHOR and DEVELOPER TEAM working group had the following members:

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PREFACE

This GCEP (PTT Global Chemical Engineering Practice) are a set of Standard Specifications that shall be used for the development of new facilities as well as improvement of existing facilities within PTT Global Chemical Public Company Limited (GC) and its subsidiaries (GC Group).

GCEP's have been developed based on GC's past experience and involvement during the design and construction of its facilities, and operation and maintenance of those facilities within the GC Group.

GCEP makes reference to International Standards and Local Regulation where applicable.

Any deviation from this GCEP either initiated by GC or by the CONTRACTOR/VENDOR is subjected to a Deviation Approval Process.

The intended USER's of these GCEP include Project Team, Plant/ Facility OWNER, Maintenance and Operations within GC as well as appointed CONTRACTOR's and VENDOR's.

This GCEP and its Specification, Drawing and/or Information is Confidential, Proprietary and Legally Privileged, and is intended for use within PTT Global Chemical Public Company Limited (GC) and its subsidiaries (GC Group) only. No recipients are allowed to disclose, distribute, copy, modify, retransmit, or disseminate this Confidential Information to any Third Party without GC's consent.

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1.0 **SCOPE**

The GCEP covers the minimum technical requirements for the engineering, design and installation for all GC facilities and shall be applied on new projects. Nevertheless, applying GCEP requirement on existing facilities in a retrofit manner is encouraged where justifiable under the facility owner's discretion.

If any requirement is intended to be applicable only to a specific individual Business Unit, it shall be clearly denoted by a capital letter coding corresponding to its Business Units as follows :-

- [R] = Refinery
- [A] = Aromatics
- [O] = Olefins
- [P] = Polymer
- [E] = Ethylene Oxide Based
- [Ph] = Phenol
- [B] = Bisphenol A
- [G] = Green

Otherwise the requirement shall be considered to be applicable to all Business Units.

Any deviation from this GCEP either initiated by GC or by the CONTRACTOR/ VENDOR is subjected to a Deviation Approval Process.

In the case that of this Specification, defines a different requirement to that of the relevant design Codes, Standards, and National/Local Regulations. The most stringent requirement shall be applied.

This Specification does not fully define application parameters and procedures for every standard practice. It is considered that specialist applicators and the paint VENDOR are familiar with these requirements. However, the general requirements and some detailed Specifications shall be defined. The surface preparation standard and Coating film thickness presented herein are the minimum requirements.

Equipment items covered by this Standard include above Ground Piping, Equipment, Un-Galvanized Structural Steelwork, Furnace Plating, Storage Tanks and other miscellaneous unprotected Steel items.

This Specification excludes; internal linings, protective Coatings applied specifically for transportation and shipment, and painted markings applied to identify Materials of construction.

This Specification does not specify paint color schemes or high temperature resistant painting Systems for operating temperatures above 650°C, nor temperature indicating paints for refractory lined Equipment items.

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2.0 **DEFINITION**

2.1 Common Definitions

GC	PTT Global Chemical Public Company Limited (GC) and its affiliates.
GCEP	PTT Global Chemical Engineering Practice
GCEP COMMITTEE	A group of GC management responsible for officially authorization of issuing GCEP.
ESO	GC Engineering Standard Office responsible for the administration of GCEP.
OWNER	The Project Owner or Plant Owner or relevant Executive at VP or higher level who have full responsibility for complying with this Specification.
Discipline Manager	Division Managers who technically authorize over GCEP contents.
Discipline Technical Committee	Engineering Progression Committee and Specialist.
Developer Team	A group of staffs who have knowledge of concerned subjects to develop and update GCEP.
TA	GC Technical Authority those are responsible for technical review and advice.
USER	Any intended Organization obligated to use this Specification for Engineering, Design, Procurement, Construction, Commissioning and Operation for the Plant/ Facility.
CONTRACTOR	Any intended Organization obligated to carry out the services of Engineering, Design, Procurement, Construction, Commissioning, Operation and/or Project Management Consultancy (PMC) for the GC facility.
VENDOR	Any intended Organization obligated to carry out the service of Supply of any Equipment and Materials or Temporary Facilities including where necessary assistance in connection with the Erection, Testing and Commissioning of any such items; the term shall also include any Person or Company that carry out the service of provision of such assistance in respect of Equipment and Materials.
SHALL	Means a mandatory requirement within this Specification. Any deviation requires shall be review by TA, prior to submitted, with their advices, to OWNER for review and approval.
SHOULD	Means an optional requirement within this Specification, OWNER shall be consulted for decision on this requirement.
[\$]	This indicator denotes being either an optional requirement,

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or additional OWNER's decision required which has to be prior resolved with OWNER and recorded in a "GCEP Resolution Record Form".

[R] This indicator denotes that the requirement in that particular paragraph is intended to be applicable for the Refinery Business Unit only.

[A] This indicator denotes that the requirement in that particular paragraph is intended to be applicable for the Aromatics Business Unit only.

[O] This indicator denotes that the requirement in that particular paragraph is intended to be applicable for the Olefin Business Unit only.

[P] This indicator denotes that the requirement in that particular paragraph is intended to be applicable for the Polymer Business Unit only.

[E] This indicator denotes that the requirement in that particular paragraph is intended to be applicable for the Ethylene Oxide Based Business Unit only.

[Ph] This indicator denotes that the requirement in that particular paragraph is intended to be applicable for the Phenol Business Unit only.

[B] This indicator denotes that the requirement in that particular paragraph is intended to be applicable for the Bis-Phenol A Business Unit only.

[G] This indicator denotes that the requirement in that particular paragraph is intended to be applicable for the Green Business Unit only.

GCEP Resolution Record Form The Form that is used to record the resolution and decision of any recommendation items denoted with [\$] in this Specification.

GCEP Deviation Form The Form that is used to record the OWNER's approval of deviation to this Specification.

Deviation Approval Process The work process that describes instruction of approval of any deviation from the GCEP. Refer to GCEP-GN-001, GCEP Governance Manual.

2.2 Specific Definitions

Shop A Manufacturer's or Fabricator's shop or plant works.

Field The permanent Site of Plant Construction.

T.U. Shall be defined as the painting of plant items that have previously been primed and/or over coated and require maintenance or a repair "touch-up" Coating.

Protective Coating System Shall be defined as the complete and total protection including surface preparation and intercoat cleaning, primer

material, undercoat and top coat Materials.

Chemical Pre-treatment

Treatment of the blast cleaned pipe surface with a chemical solution (e.g phosphate), before application of the primer to improve the adhesion of the coating to the substrate.

Shop Primer

Thin protective Coating normally applied for protection during transport and storage

Stripe Coat

Supplementary coat applied to ensure adequate protection of critical areas like edges, welds etc.

Cyclic service

Service in which pressure Equipment is subject to frequent fluctuations in pressure and/or temperature, Mechanical vibration or external loading that may lead to damage from fatigue.

Feathered

Gradual taper in thickness from a coated surface to an uncoated surface

Adhesive

An intermediate coating layer to improve adhesion between the primer and the coating.

Holiday

Discontinuity in a Coating, which exhibits electrical conductivity, when exposed to a specific voltage

2.3 Abbreviations

DFT

Dried Film Thickness

ASTM

American Society for Testing and Materials

ISO

International Organization for Standard

SSPC

American Steel Structures Painting Council Standards

NACE

National Association of Corrosion Engineers

3.0 ASSOCIATED CODES AND STANDARDS

The listed Codes and Standards below shall form an integral part of the requirements of this Specification. The latest Edition or/ Revision or/ Issue date of these Codes and Standards shall be used if it is not specified.

Standard No.

Standard Title

ASTM

ASTM A123

Zinc (Hot-Dip Galvanised) Coatings on Iron and Steel

ASTM A153

Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM C633

Standard Test Method for Adhesion or Cohesive Strength of Flame-Sprayed Coatings

ASTM D1210

Standard Test Methods for Fineness of Dispersion of Pigment-Vehicle Systems by Hegman-Type Gage

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ASTM D4285	Standard Test Methods for Indicating Oil or Water in Compressed Air
ASTM D4752	Measuring MEK Resistance of Ethyl Silicate (Inorganic) Zinc Rich Primers by Solvent Rub
ASTM D4940	Standard Test Method for Conductimetric Analysis of Water Soluble Ionic Contamination of Blasting Abrasives.
ASTM D520	Standard Specification for Zinc Dust Pigment
<u>ISO</u>	
ISO 12944	Paints & Varnishes – Corrosion Protection of Steel Structures by Protective Paint Systems
ISO 4624	Paints and Varnishes – Pull-Off Test for Adhesion
ISO 8501-1	Preparation of Steel Substrates Before Application of Paints and Related Products – Visual Assessment of Surface Cleanliness
ISO 8502-3	Preparation of Steel Substrates Before Application of Paints and Related Products – Test for the Assessment of Surface Cleanliness – Part 3: Assessment of Dust on Steel Surfaces Prepared for Painting (Pressure Sensitive Tape Method)
ISO 8502-6	Preparation of Steel Substrates Before Application of Paints and Related Products – Test for the Assessment of Surface Cleanliness – Sampling of Soluble Impurities on Surfaces to be Painted – The Bresle Method
ISO 8502-9	Preparation of Steel substrates before application of paints and related products -- Tests for the assessment of surface cleanliness -- Part 9: Field method for the conductometric determination of water-soluble salts
ISO 8503	Preparation of Steel Substrates Before Application of Paint and Related Products
ISO 11124	Preparation of Steel substrates before application of paints and related products -- Specifications for metallic blast-cleaning abrasives -- Part 1: General introduction and classification
ISO 11126	Preparation of Steel substrates before application of paints and related products -- Specifications for non-metallic blast-cleaning abrasives -- Part 6: Iron furnace slag
ISO 11127	Preparation of Steel substrates before application of paints and related products -- Test

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methods for non-metallic blast-cleaning
abrasives -- Part 1: Sampling

ISO 19840	Paints and varnishes -- Corrosion protection of Steel Structures by protective paint Systems -- Measurement of, and acceptance criteria for, the thickness of dry films on rough surfaces
ISO 1461	Hot-dip Galvanized Coatings on fabricated iron and Steel articles – Specifications and test methods.
ISO 2409	Paints and Varnishes- Cross-Cut Test
ISO 2808	Paints and Varnishes- Determination of film thickness
ISO D5402	Standard Practice for Assessing the Solvent Resistance of Organic Coatings Using Solvent Rubs

SSPC/NACE

SSPC - PA1	Shop, Field and Maintenance Painting
SSPC - PA2	Measurement of Dry Paint Thickness with Magnetic Gauges
SSPC - SP1	Solvent Cleaning
SSPC - SP2	Hand Tool Cleaning
SSPC - SP3	Power Tool Cleaning
SSPC - SP5/NACE No.1	White Metal Blast Cleaning
SSPC - SP6	Commercial Blast Cleaning
SSPC - SP7	Brush-off Blast Cleaning
SSPC - SP8	Pickling
SSPC - SP10/NACE No.2	Near White Metal Blast Cleaning
SSPC - SP11	Power Tool Cleaning to Bare Metal
SSPC - AB 2	Abrasive Specification No.2, Cleanliness of Recycled Ferrous Metallic Abrasives
SSPC - AB 3	Newly Manufactured or Re-Manufactured Steel Abrasive
SSPC - VIS 1	Guide and Reference Photographs for Steel Surface Prepared by Dry Abrasive Blast Cleaning
SSPC - VIS 3	Guide and Reference Photographs for Steel Surface Prepared by Power-Tool and Hand-Tool

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	Cleaning
SSPC Paint 20	Zinc-Rich Coating, Type I-Inorganic and Type II-Organic
NACE SP 0188	Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates.

4.0 **EXECUTION OF THE WORK**

4.1 The following surfaces and Materials shall require painting.

- a. Carbon Steel
- b. Low Alloyed Steel
- c. Austenitic Stainless Steel
- d. Fiber reinforced plastic (FRP) & Glass reinforced epoxy (GRE)

4.2 The following surfaces and Materials shall not require painting unless specified by Project.

- a. Non-Ferrous Materials
- b. Foundations, Concrete Structures Concrete or Mortar Fireproofing.
- c. Plastic and/or Plastic Coated Materials.
- d. Flange Faces (Raised Face, Grooved Face, etc.).
- e. The contact surface of an earthing System.
- f. "In-Line" Instruments (painted by Vendor)
- g. Fasteners
- h. Insulation Jackets
- i. Any Equipment furnished completely primed and finish coated that suitable for design temperature by the Equipment VENDOR which approved by OWNER. (Pumps, Motors, valves and other Machinery, Instruments, Instrument boards etc.).
- j. Asbestos free Cement Sheeting, Slate, Brick, Tile, Glass and similar weather resistant Materials.
- k. Buried Steel Piping that wrapped.
- l. Machined surfaces of Machinery and gasket contact surfaces.
- m. Tripping mechanisms of Steam Turbines, Valve Stems, Movable Linkage on Equipment, Grease-Fittings, Compressor or Pump Shafts or any such surfaces on Equipment that are normally lubricated or have close working tolerances.

4.3 The items listed below shall be shielded and protected to prevent damage during surface preparation and paint material application operations. All openings, including those that are flanged or threaded shall be sealed to prevent entry of blast abrasive or paint material. After completion of painting operations all material used for shielding and sealing shall be removed unless instructed otherwise:-

- a. Nameplates

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- b. Packing Glands and Seals
- c. Bearings
- d. Rotating Equipment Couplings and Shafts
- e. Lubrication Fittings
- f. Pressure Gauges
- g. Gauge and Flow Indicator Glasses
- h. Motor Starters
- i. Instrument Dials
- j. Exposed Linkages
- k. Valve Stems
- l. Light Bulbs, Enclosures and Reflectors
- m. Air Intakes
- n. Rubber Parts and Plastic
- o. Valve Trim Discs.

4.4 Surfaces requiring to be protected by wire mesh guard shall be fully painted using the same paint system as un-insulated surface.

Surfaces requiring to be protected by insulation for Personnel Protection shall be fully painted using the same paint system as insulated surface.

4.5 CONTRACTOR shall propose the methods to protect the thread of Valves' Stems and Bolts and Spectacle Blinds.

5.0 **DESIGN REQUIREMENTS**

5.1 Painting system

- a. For each project / contract, paint manufacturer / vendor shall submit product specification which applicable with paint system in the Table 1.1 – 1.3 of this GCEP for review and approve. Technical review shall be approved by TA2.
- b. Maximum hot application temperature shall be submitted for maintenance and minor repair painting system.
- c. Product datasheet shall be identified essential information for technical review such as minimum surface preparation, service temperature, **maximum hot application temperature**, and etc. If essential information is missing, the warranty letter shall be issued from paint manufacturer.
- d. VENDOR shall specify dry film thickness of each painting system to be used and it shall comply with product datasheet. The dry film thickness shall specify in range of minimum to maximum.
- e. Lead containing paints, coal-tar epoxy paints and coal-tar urethane paints shall not be used because of the associated health and environmental restrictions that apply.
- f. At the site of construction, the Mechanical CONTRACTOR shall be responsible for properly handling and storing the primed and painted items prior to erection to

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minimize damage to primed and painted surfaces.

- g. The protective Coating System applied by the VENDOR to proprietary items such as Motors, Pumps, Turbines, Instrument panels etc. shall be suitable for the intended operational and environmental conditions at the site of construction and give a service life equal to, or better than the equivalent System contained within this Specification.

- h. The selected protective paint Systems shall generally be based on normal Design temperatures including any cyclic, intermittent or dual operating conditions of pipe work or Equipment items with the following exceptions :

Items internally lined where paint Systems shall be selected to suit the anticipated outside wall temperatures.

- i. Insulated Carbon Steel Piping shall have a full paint System applied under the Insulation according to the Systems specified for the applicable service temperature.

- j. Painting system selection to suit actual using condition where Maximum and Minimum design temperature are specified on design condition.

When normal operating temperature is 0 degC and above, critical factor for paint system selection shall be maximum design temperature. Therefore, maximum design temperature shall be the basis for paint system selection.

When normal operating temperature is below 0 degC, critical factor for paint system selection shall be minimum design temperature. Therefore, minimum design temperature shall be the basis for paint system selection.

Where there is Cyclic Temperature or Heating up temperature such as steam out, the painting system shall be chosen to cover such condition.

- k. At the commencement in advance of the work commencing, all the selected Systems shall be applied on the Test Reference Panels representative of the structure to be coated. A separate test reference panel shall be prepared for each combination of surface preparation method and Coating application method to be used (e.g. spray applied to abrasive blasted surface, brush applied to Power Tool prepared surface).

The System shall be applied in the presence of the VENDOR, the Sub-CONTRACTOR/Module Fabricator and the OWNER/CONTRACTOR's representative, in climatic conditions that are as similar as possible to worst case conditions that can be foreseen during the work in the construction site. Application shall be performed by the Coating applicator that will be responsible for application during work. After approval, shall be used as the standard for the remainder of the Project work.

- l. In addition to the requirements of this Specification, the paint VENDOR's instructions shall be carefully followed. In case of conflict, the most stringent requirement shall govern.

- m. If painting works are done at shop, before transportation shall be the painting with full system consist as Primer, the Intermediate and the Top coat. These precautions to avoid deteriorate of Primer and the Intermediate by exposure to environment as UV, Moisture, Dust and Salts contaminate.

- n. Any Equipment (Pumps, Motors, valves and other Machinery, Instruments, Instrument boards) furnished completely primed and finish coated which

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could not identified the paint system, it shall be assumed as "Painting system CB-4 (carbon and low allow steel) or SB-4 (stainless steel)".

5.2 Protrusion Through Insulation Systems

The protrusions through Insulation Systems such as Saddles, Nozzles, Manways, Davits, and Miscellaneous Clips and Brackets will normally be coated the same as the item to which it is welded.

5.3 Under Concrete fireproofing

Steel structure(s) under concrete fireproofing shall be coated with coating system K. Steel structure member(s) that require fireproofing (full or partial) able to apply the coating system K entire member(s), the protrusions through fireproofing shall be add 2 components aliphatic polyurethane at least 50 µm for top coating. Coating is not required for galvanized surface(s) under fireproofing.

Equipment's Skirt(s) under fireproofing for carbon and low alloyed steel shall be coated with the Insulated coating system of equipment design temperature.

The protrusions through fireproofing such as Saddles, Nozzles, Manways, Davits, and Miscellaneous Clips and Brackets will normally be coated the same as the item to which it is welded.

5.4 Precautions for Austenitic Stainless Steel and High Alloys Materials

- a. Special care shall be taken for the protection of Stainless Steel and high Alloys Materials from blasting, Overspray and Coatings intended for Carbon Steel Materials, particularly from Coatings containing Zinc.
- b. Austenitic Stainless Steel surfaces which are to be coated shall have all Dust, Dirt and loose contamination removed by washing with condensate, distilled or other Chloride free water. Hard to remove dirt may be removed with clean Stainless Steel wire brushes which have not been used on Carbon Steel surfaces. Oil and grease shall be removed by wiping with clean rags soaked in clean, uncontaminated Aromatic Hydrocarbon solvents, Ketones or Alcohols. Aliphatic Hydrocarbon and Chlorinated solvents shall not be used. Alternately, high pressure steam cleaning may be used to remove oil, grease and other surface contaminants.
- c. The Power Tool and Hand tools used to clean Carbon Steel shall not be used to clean Stainless Steel and Non-Ferrous Metals.
- d. In prevention of galvanic corrosion, paint shall be applied at least 300 mm from the dissimilar joint toward the uncoated material.

5.5 Equipment

Equipment shall be shop coated to the maximum extent possible. When not fireproofed, the equipment's skirt(s) both internal and external surfaces will normally be coated the same as Equipment coating system.

5.6 Machinery and Packaged Equipment

Completely fabricated Equipment, Machinery, etc., shall be shop primed and finish painted whenever practical. Field painting shall be kept to a minimum.

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5.7 Safety Color Coding

The safety color coding scheme shall be in accordance with ANSI A31.1 1981. Unless specified otherwise, the following items shall be color coded:

- a. The location of first aid equipment shall be marked safety green. Columns having first aid equipment shall be colored green on all sides to a height of 3 m.
- b. The location of fire extinguishers shall be colored safety red. Columns having fire extinguishers shall be coated red on all sides to a height of 3 m.

6.0 SURFACE PREPARATION

- 6.1 The fabricator shall remove all welding defects and surface discontinuities such as Lamination, Surface Cracking from welded areas and grind out any sharp edges giving a minimum radius of 2 mm. Prior to painting the welds and surrounding area shall be evaluated using SSPC VIS 1 or SSPC VIS 3. Preparation shall include the cleaning of the chamfered paint edges and the removal of any burnt or scorched paint.
- 6.2 Prior to blast cleaning all grease and oil contamination shall be removed by wiping or scrubbing the surface with rags or brushes wetted with solvent followed by a wipe down with clean dry cloths. Alternatively, proprietary emulsifying agents may be used; where this method of cleansing is employed the surface shall be thoroughly washed with clean water and then dried before proceeding.
- 6.3 Blast cleaning shall not be carried out whilst the temperature of metal surfaces are less than 3°C above the dew point of the surrounding air or when the Relative Humidity of the air is greater than 85%. However, the Product Data Sheet of the Paint VENDOR shall be followed if requires a more stringent maximum Relative Humidity.
- 6.4 Abrasive blasting (Whip) blasting shall only be performed in daylight or under artificial lighting that is equivalent to daylight and free of significant shadows.
- 6.5 All blast cleaned surfaces shall be primed before visible oxidation occurs within four hours of the commencement of blasting. If visible re-rusting occurs after more than two hours have elapsed then the metal surfaces shall be lightly re-blasted and where necessary washed to remove rust blooming and accumulation of soluble salts. Spent blast abrasive shall be completely removed from the prepared surface by either vacuum cleaning or stiff-brush sweeping.
- 6.6 The cleanliness of blasted surfaces exposed to ambient condition shall be as per ISO 8501-1 SA2.5 or SSPC –SP10/NACE No.2 Near white metal blast cleaning. For surfaces to be immersed or buried and the internals of production vessels, the abrasive blasted surface shall be as per ISO 8501-1 SA3 or SSPC –SP5/NACE No.1 White metal blast cleaning prior to application of Coating.

The cleanliness of blasted surfaces exposed to ambient condition shall be as per ISO 8501-1 SA1 or SSPC –SP7/NACE No.4 Brush-off blast cleaning for stainless steel.

6.7 The abrasive blast shall be selected as first priority, the power tool to bare metal SSPC – SP11 could be selected for below:

- a. Field joint
- b. Limited access for blasting

Or other applications where agree with OWNER. The paint system shall be complied with table 1.2 of GCEP-PP-005 where SSPC – SP11 paint system applied with surface profile is not less than 25 microns.

6.8 The CONTRACTOR shall select an abrasive blast media that ensures a consistent sharp and angular. Surface roughness shall be in accordance with recommended by VENDOR for each product. Where Surface roughness is not specified by a VENDOR, the general guidance surface profile values as noted below.

Surface		Anchor Pattern, μm
Steel	Coating Total DFT < 500 μm	40-60
	Coating Total DFT 500-1000 μm	60-100
	Coating Total DFT > 1000 μm	90-110
Aluminium		20-30
Hot Dipped Galvanizing		20-30
Stainless Steel		30-40

6.9 The roughness of blast cleaned surface is reproduced using a replica tape, such as “Testex” and measured with a calibrated Testex Dial gauge such replica tapes shall be retained and attached to the daily paint report as a permanent record. Any other alternative method shall be submitted for TA approval.

6.10 The frequency to check the surface profile shall be minimum of 2 samples per shift. However, if one shift has produced a blasted area of more than 100 m², the frequency shall be 1 sample per 100 m².

6.11 Blast cleaning shall not be carried out in the vicinity of painting works and dry grit blasting shall not take place when wind-blown dust and debris could contaminate uncured Coatings. Other contaminants such as insect swarms or falling debris should be considered prior to beginning Coating work.

6.12 Dust, blast abrasives etc. shall be removed from the surface after blast cleaning such that the particle quantity and particle size do not exceed rating 2 of ISO 8502-3. Frequency shall be minimum of 1 sample per shift.

6.13 The testing for salt contamination shall be performed immediately after abrasive blasting of Carbon Steel, Low Alloys Steel and Stainless Steel items in accordance with ISO 8502-6 (Bresle) and ISO 8502-9 (Conductometric). Frequency shall be minimum of 1 sample per shift. However, if one shift has produced a blasted area of more than 500 m², the frequency shall be 1 sample per 500 m². The maximum

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allowable level of Total Salt contamination shall not be over 25 mg/m² for carbon and low alloyed steel and not be over 20 mg/m² for stainless steel.

7.0 **FIELD PREPARATION**

- 7.1 All the provisions for Shop Surface Preparation shall be applicable for Field Preparation. If the specified requirements are impractical because of Field Conditions, this Specification becomes invalidated and revised Coating Systems shall be devised to suit the imposed Field Preparation conditions.
- 7.2 Any revisions to –or, deviations from- this Specification shall only be acceptable when approved in writing by TA, written instruction shall be given prior to any revised Coating related activities being executed.
- 7.3 Shop prepared and primed surfaces shall be field repaired after erection of primed plant.
- 7.4 Over areas considered to be inaccessible after erection, shop-applied primer repairs, field-applied primers, undercoats and finishing coats shall be applied, dried and cured before construction takes place. Care shall be taken to allow sufficient drying and curing time before moving, lifting or erecting newly painted surfaces.
- 7.5 The cleaning methods undertaken shall remove all welds flux slag and fumes, welds heat oxides and welds metal spatter in addition to all loose mill scale or loose paint if present after welding and testing has been completed.
- 7.6 Prior to Mechanical preparation and application of primer paint the metal surface and feathered area of shop Coating shall be solvent cleaned and thoroughly washed with clean water to remove all soluble salts and dried in a manner which will not re-deposit contaminants on to the cleaned surface. The feathered area is to overlap to the existing Coating by at least 50 mm. (2 in).
- 7.7 Field surface preparation local to welds areas shall be carried out by the vacuum blast method as far as is practicable within the field construction conditions. Where surface cleaning by abrasion blasting is prohibited, welds and other small areas shall be prepared by use of Power Tool Cleaning or Hand Tool Cleaning or a combination of these methods depending on the contour and position of the surface to be painted and the approved primer material suitable for the cleaning method selected.
- 7.8 Surface preparation with Power Tools shall achieve a level of Cleanliness equivalent to SSPC SP11 and Hand Tools shall be achieving a level of Cleanliness equivalent to SSPC SP2.
- 7.9 Power Tool cleaning is acceptable in areas where abrasive blasting is not suitable and where the tools and techniques used provide a sharp, angular profile of the specified depth and level of cleanliness. CONTRACTOR shall submit for TA approval any Power Tool to be used, list its Name, Model, and Manufacturer.
- 7.10 If the Power Tool used is air-powered, then the area must be solvent wiped to ensure that the tool did not contaminate the surface with oil through worn seals.
- 7.11 The use of grinding discs may be included to achieve the removal of tightly adhering welds splatter, but great care shall be taken to ensure that surfaces shall not be polished nor left with a detrimental roughness.
- 7.12 Shop applied paint shall be feathered back to allow the correct application and build-up of required site applied paint Coatings.

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8.0 **ABRASIVE BLAST**

- 8.1 Only dry blast cleaning techniques shall be employed. Sand is not permitted.
- 8.2 When Steel Grit and Shot abrasive is used, the minimum grit/shot ratio shall be 65%. Recycled abrasive may be used only in the blast rooms where automated dust removal Equipment exists.
- 8.3 A non-metallic blast abrasive, i.e. Aluminium Oxide or Garnet, shall be used when abrading Stainless Steel surfaces.
- 8.4 Abrasive size shall be selected to best suit the type of priming material and to satisfy the paint VENDORS recommendation, bearing in mind the restriction imposed above.
- 8.5 Recycled Ferrous metallic abrasives shall be performed "vial test" to check the cleanliness and contaminate of residual painting, mill scale, rust and other foreign matter in accordance with SSPC AB2 Cleanliness of Recycling Ferrous Metallic Abrasive and SSPC AB3 Newly Manufactured or Re-Manufactured Steel at the start of each shift.
- 8.6 Wet abrasive blasting, where necessary, may be used only after approval in writing by TA.

9.0 **COMPRESSED AIR**

- 9.1 Compressed air supply used for blast cleaning shall be free from water and oil. Adequate oil & moisture Separators and Traps shall be provided and installed in the coolest part of the System. They shall be regularly emptied to prevent carry-over of water and oil. Accumulations of oil and moisture shall be removed from the Air Receiver by regular purging. The Air Compressor shall not deliver air at a temperature in excess of 110°C.
- 9.2 Nozzles shall be discarded and replaced when 50% worn.
- 9.3 Where air-operated Equipment is used, the operator's hood or head gear shall be ventilated by clean cool air served through a regulator filter, to prevent blast cleaning residues from being inhaled.
- 9.4 Every 4 hours the moisture content at the blast nozzle shall be checked to ensure that the Air Aftercoolers are working adequately and eliminating any moisture. However, this requirement may be waived if no non-compliances are detected at the Blast Nozzle. To do this per ASTM D4285 the abrasive material Ball Valve on the Blast Pot shall be closed and the blast pot turned on so that the air (and not the abrasive slag) is expelled out to the blast nozzle and on to a piece of white paper or cloth. Then expel the air onto the flat Steel surface to check the condition of air being expelled for moisture content. If the moisture flow cannot be contained, an additional moisture filter will be required for controlling moisture as per Specification requirement.

10.0 **COATING APPLICATION**

10.1 Application Condition

- a. Coating application shall only be performed by competent applicators during daylight hours or under artificial lighting that is equivalent to daylight and free of significant shadows.

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- b. Conventional solvent based paints shall not be applied to exterior surfaces in damp, humid weather when air temperature is below 5°C or when the metal surface temperature is less than 3°C above the dew-point temperature.
- c. Chemically cured paints shall not be applied at air or metal surface temperature below 10°C or when Relative Humidity is above 85%. However the Product Data Sheet of the Paint VENDOR shall be followed if requires a more stringent maximum Relative Humidity.
- d. Before the application of paint Materials the VENDORS recommendations regarding temperatures and ambient conditions of application shall be consulted and reviewed for possible special application requirements.
- e. The maximum hot application temperature shall be followed the product datasheet or recommendation letter from manufacturer. If it's not specified, paint materials shall not be applied over metal surfaces having temperatures exceeding 38°C.
- f. Paint Materials shall not be applied in rain, snow, fog or mist, nor to wet or damp surfaces. Paint shall not be applied to frost or ice coated surfaces.
- g. All painting applied shall consist of good coverage and good leveling without sagging on vertical surfaces. In general, curing of paints shall provide optimum smoothing of the paint film.

10.2 Paint Mixing

- a. Paint material shall be mixed by Mechanical stirrers, paddle mixers, can vibrators or can shakers. Partial mixing is not allowed between part A and part b.
- b. Constant Mechanical agitation shall be maintained in the paint reservoir throughout paint spraying operations.
- c. Thinners shall only be added strictly in accordance with the paint VENDORS' Instructions.
- d. Thinners shall not cause the total volatile organic compound (VOC) content to exceed Local Regulations and thinners shall only be added strictly in accordance with the Paint VENDOR' Instructions. The type of thinners shall also comply with the Paint VENDOR' Instructions.
- e. When thinners are permissible, they shall be added during the paint mixing process.
- f. Paint components shall be mixed in the proper ratio as supplied by the paint VENDOR in proportional containers. Mixing of "guessed" quantities is not permitted. Where required, the Paint VENDOR indicated induction time shall be observed. Mixed paint exceeding the pot life shall not be used.

10.3 Shop and Field Painting

- a. Shop applied Coatings shall be dried and completely cured to permit transport to the site of construction without damage
- b. Shop primed surfaces contaminated during transport and site storage shall be thoroughly washed down with fresh clean water and allowed to dry before the application of further paint Coatings.
- c. If the coated surface is burnt or damaged from welding/fabrication it shall be

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removed from an area about 50 mm wide either side of the line of burning or welding. The remainder of the paint shall be chamfered back for approximately a further 100 mm to 150 mm.

- d. Aluminium paint shall not be applied to any Steel or Iron surfaces likely to attain a temperature greater than 100°C in any building where inflammable vapour may be present.
- e. Inorganic Zinc Primer, and other primers when required by VENDOR, shall be applied using an agitated pot.
- f. Inorganic Zinc Primer shall have dry spray removed with a wire screen and shall be confirmed as sufficiently cured according to ASTM D 4752 the Methyl Ethyl Ketone (MEK) rub test per Coating VENDOR's Instructions before top coating. Frequency should be a minimum 3 tests per 500 m2 and the results shall have an average resistance rating of not less than 4.
- g. Each coat of a multi-coating materials shall be a contrasting color to indicate the extent of coverage.
- h. Each coat of a multi-coat System shall be allowed to cure sufficiently in accordance with Paint VENDOR's Instructions prior to top coating. Minimum cure time between coats shall be in compliance with VENDOR's Instructions for the time-averaged substrate temperature and Humidity.

10.4 Application Methods

- a. Paint Materials shall only be applied by the Paint VENDORS recommended method. Care shall be taken to select paint Materials to suit the intended application method.
- b. Particular attention shall be paid to the preparation and painting of corners, Edges, Welds, Rivet Heads, Small Brackets, Nuts, Interstices and hard-to-reach areas. Prior to applying each coat of the Coating System, these positions shall be pre-coated by Brush, i.e. 'Stripe Coated' (using a contrasting shade to the main coat where possible) to ensure specified nominal dried film thicknesses are achieved.
- c. Spray Guns shall be held perpendicular to the surface being coated at a maximum distance of 300 mm from the surface and no closer than 200mm to the surface. Wet film thickness (comb) gauge shall be used by paint spray gun operatives to ensure an even and adequate DFT is achieved.
- d. Material and air pressure on the Spray Guns shall be regulated to achieve optimum atomization of the paint with the lowest possible pressures to avoid spray ricochet from the surface being painted.
- e. Overlapping horizontal passes shall be made for each spray coat to obtain uniform application. Particular care shall be exercised to ensure that a uniform, complete wet coat is applied. Semi-dry spraying results in poor flow-out with excessive pin holing and shall be avoided.
- f. Coats of paint shall be uniformly applied without sags, runs, spots, damage or contamination; such irregularities shall be removed and the affected area recoated.
- g. Each coat of paint shall be in a proper state of cure or dryness before the application of the succeeding coat.

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- h. Coatings which have weathered for more than six (6) months shall be thoroughly inspected prior to application of further coats to prevent intercoat adhesion problems.
- i. To minimize intercoat contamination, succeeding coats shall be applied with a minimum time lapse consistent with proper drying time, work and erection schedules.
- j. Roller application of the first primer coat is not acceptable. Rollers when used shall be manufactured from good quality 'carpet pile' or lamb's wool.
- k. When paints are applied by brushing, brushes shall be of a style and quality that will permit proper application of the material. Brushing shall be done so that a smooth coat as nearly uniform in thickness as possible is obtained. There shall be no deep "detrimental" brush marks.
- l. The overcoating interval time for each layer of painting System shall be in accordance with the Paint VENDOR Data Sheet.
- m. When Carbon Steel Pipe is welded/ connected with Stainless Steel Joints, the Coating System for Stainless Steel shall be applied both sides of the joint and the coated overlap into the Carbon Steel shall be minimum 50 mm.

11.0 APPLICATION RESPONSIBILITY

11.1 General

- a. Certificates of release, confirming that successful completion of all appropriate tests and safety checks have been carried out to the released working areas, shall be obtained before the commencement of painting work.
- b. All provisions of this Specification shall apply to both Shop and Field painting.
- c. All paint work shall be carried out in a neat, thorough and workmanlike manner.
- d. Wet paint shall be protected against damage by the Painting CONTRACTOR.
- e. All shipping tags, wires, strings and other means of temporary or shipping identification on surfaces to be painted shall be removed only after checking with and receiving authorization from the CONTRACTOR's Representative.
- f. The Painting CONTRACTOR shall be responsible for protection against damage to abutting property, vehicles, pedestrians and other portions of the structure, due to the painting operations.
- g. All Instruments, Gauge Glasses, etc. (see 4.3) be protected from paint splatter by the Painting CONTRACTOR.
- h. The job site shall be maintained clean and neat during the paint application work.
- i. Paints, thinners, solvents and rags, waste etc., soiled with these Materials shall be kept in tightly closed containers whilst on the job site and not in use.
- j. The CONTRACTOR/OWNER/PMC's Representative or his designate shall have the right to stop the application of paint when in his opinion; conditions may result in injury to the applied Coatings.
- k. The paint applicator shall provide a temporary shelter for storage and mixing of paint Materials.

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- I. During both application and drying, adequate ventilation shall be provided if the work area is enclosed.

11.2 Safety Requirements

- a. The Paint CONTRACTOR shall be acquainted with safety and security requirements for Equipment and Materials used.
- b. The Paint CONTRACTOR shall observe and carry out all work in strict compliance with Plant, Local and National Health and Safety Rules and shall obtain all necessary Permits from the applicable Authorities.
- c. The paint CONTRACTOR shall be responsible for earthing all his Mechanical Equipment to prevent build-up of static electricity.
- d. Before painters, blasters and general operatives are sent to work in any Equipment area; the working environment should be studied to detect any hazards such as concentration of noxious fumes, presence of splash or spillage of harmful liquids, conditions of excessive heat, or areas where oxygen content of the air may be dangerously low. Where these conditions exist, they must be corrected before painting personnel are permitted to enter the area, or painting personnel must be protected by Materials or devices that will allow them to work without fear of injury or illness that might otherwise result from exposure.
- e. The following Safety Checklist identifies the minimum requirements for the Paint CONTRACTOR to ensure the welfare of his workforce and any other workers in the surrounding area. The Paint CONTRACTOR is responsible for producing a fully comprehensive list.
- f. The quantity of Volatile Organic Compound (VOC), in grams per liter, (pounds per gallon), shall be in compliance with the legal local, national and international values.

12.0 INSPECTION REQUIREMENT

12.1 Qualification of Inspectors.

Personnel carrying out inspection or verification shall be qualified in accordance with NACE Coating Inspector or approved equivalent. Assigned coating inspector shall have competency and proficiency to perform destructive testing according to the GCEP requirement.

- 12.2 Before painting activities commence the CONTRACTOR shall submit an Inspection Procedure Report indicating Inspections and Tests to be conducted during preparation and application of the paint System for approval.

- 12.3 Wet Film Thickness (WFT) shall be carried out by spot checks during the painting operation to ensure that film thickness is being maintained. These shall be performed accordance with ISO 2808, Method No.1A- Comb gauge.

- 12.4 Dry Film Thickness (DFT) Testing shall be performed in accordance with ISO 2808, methods 7B, 7C and 7D. The Coating thickness gauge shall be calibrated in accordance with the procedure detailed in SSPC-PA2 using Electronic Gages (Type 2).

The allowable thickness shall be within specific dry film thickness (DFT) range which provide by paint manufacturer or VENDOR. If the dry film thickness was found thicker than the specified DFT range, it shall be blasted and re-painted to obtain

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specified thickness. If the dry film thickness was found thinner than the specified DFT range, it shall be painted to obtain specified thickness.

12.5 Film thickness measuring shall not include thickness of pre-fabricating primers. In the absence of pre-fabricating primer film thickness records, 25 microns shall be allowed for pre-fabrication primers whilst checking further film coat thicknesses.

12.6 The adhesion of the primer to the Steel substrate and the intermediate coat adhesion of subsequent coats after curing shall be determined by the application of either a cross-cut test in accordance with ISO2409 or a pull-off test described in ISO4624.

Adhesion testing shall be carried out at least one per Equipment or 100m² of complete coated surface by Coating System on properly cured Coating. For all Coating Systems, Coating found with adhesive/ cohesive failure below 5 MPa shall be considered nonconforming.

12.7 All phases of cleaning and painting and galvanizing operations of all shop and field painting shall be subject to spot check inspection by the CONTRACTOR's and appointed Representative.

12.8 This periodic inspection will be made to ensure that the paint applicator is accurately following the recommendations of this Specification and the Paint VENDOR's Instructions.

12.9 Steelwork that will be exposed to continual splash or full immersion of liquids including chemicals, salt water, wash down water etc. shall be fully tested for holidays and pinholes. Other Steelwork shall be tested for holidays and pinholes at the discretion of TA or Representative should there be a concern regarding Coating integrity. Continuity "Holiday" Testing shall be in accordance with NACE SP0188 Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates. If the DFT of Coating is more than 500 µm shall be use the high voltage technique.

12.10 The responsibility for the total cleaning, painting and galvanizing operations remain entirely with the applicators who shall fully guarantee material and workmanship for the entire protective System. Inspection, approval or comments by The CONTRACTOR, their agents or by independent Inspectors shall not relieve the applicator of this responsibility.

12.11 The Paint VENDOR must provide a Technical Service Representative during the painting work.

12.12 The CONTRACTOR shall ensure that all work is inspected and appropriate testing carried out progressively to ensure that it is in accordance with the specified requirements. Inspection and testing records shall be maintained on a daily basis. As a minimum, these reports shall include appropriate information concerning the following:

- a. Job identification;
- b. Surface preparation performed and Equipment used;
- c. Surface profile measurement;
- d. Coating applied and Equipment used;
- e. Material used, batch number, color, mix ratio and thinner;
- f. Application details including which coat was applied, wet film thickness and dry film thickness of each coat;

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- g. Weather conditions including air temperature, surface temperature, Relative Humidity, dew point, wind conditions and general weather conditions. These measurements shall be taken immediately prior to surface preparation and application work, and then every three hours, or whenever a change in the weather conditions is apparent.

In addition, the CONTRACTORs facility and inspection records shall be made available to TA or Representative for inspection of the works at any time during the performance of the contract.

13.0 REPAIR OF DAMAGED PAINT SYSTEMS

- 13.1 If Coating damage has occurred exposing Steel, then the area shall be spot-blast cleaned (Vacuum blast is acceptable). However, the repair shall be performed using the same surface preparation and Coating procedure as for production Coatings.
- 13.2 Power Tool cleaning shall only be performed to small sections using a Bristle Blaster where abrasive blasting cannot be used. The condition of SSPC SP11 shall apply to all Power Tool cleaning in accordance with SSPC-VIS 3. The resultant standard of cleanliness shall be bright clean Steel with no residual mill-scale or corrosion products and a surface profile not in excess of 25 µm.
- 13.3 Power Tool cleaning shall not be used on Structures intended for Vessel Linings, high temperature service, buried and immersion service conditions.
- 13.4 When the repair of a damaged Coating shall be smooth and sound, any repair area shall be “feathered” by Power Sanding to create a smooth transition exposing at least 50 mm over each sub-coat.
- 13.5 Areas of paint on which welds spatter, Concrete or other adherent matter has fallen shall be cleaned or washed free of the adherent material immediately and any repair or making good of the affected surface to its original standard shall be carried out before the area is over-coated with paint.
- 13.6 Where epoxy Coatings are damaged, suitable repair material supplied by the VENDOR of the original Coating shall be applied in accordance with the VENDOR's Instructions.
- 13.7 All repairs shall be subject to Inspection.

14.0 Galvanized ITEMS

14.1 Requirements

- a. For items which are hot-dip Galvanized, ASTM Specification A123 as appropriate shall apply in terms of Coating thickness.
- b. Articles specified to be Galvanized shall be completely fabricated prior to the application of the hot dip galvanizing process.
- c. Items shall not be hot dipped before surface preparation has been carried out in a full and correct manner.
- d. Surfaces shall be free of old paint, oil, grease, welds, slag deposits and laminations; and rolling fabrication defects eliminated prior to hot dipping.
- e. The thickness of Galvanizing shall conform to Table 1 in ASTM A123.

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- f. Fasteners, the Iron and Steel Hardware shall meet the requirements of ASTM A153.
- g. All Equipment platforms, Stair ways, ladders, ladder Cages, stair trades, safety gates and handrails shall be hot-dip galvanized.
- h. Ladders, ladder cages, safety gates and handrails shall be coated by coating system I after galvanizing. Top coat color shall be in accordance with Table 2 of this specification.

14.2 Repairs

- a. Surfaces requiring spot repair shall be have a surface preparation in accordance with SSPC SP3 Power Tool cleaned.
- b. For touch-up of drilled and/ or cut edge damaged areas of Galvanized Steel, surfaces shall be Power Tool cleaned and painted with one coat of Two Component High Build, Surface Tolerant Epoxy to a nominal DFT of 100 microns. Major damage shall be re-Galvanized in accordance with the above mentioned Standard.

15.0 Coating Materials

15.1 General

- a. Coating Materials shall be obtained from OWNER approved Paint VENDORS using the full generic description given in the table of this Specification. The Painting CONTRACTOR shall submit to the CONTRACTOR the name of VENDORS and details of their material products to be used, for CONTRTACTOR review and agreement.
- b. All Coating Material shall be supplied in the VENDOR's original cans, durably and legibly marked with the description of the contents. This shall include the Specification number, the color reference number, the method of application for which it is intended, the batch number, date of manufacture and the VENDOR's name, initials, or recognized trade mark. No intermixing of different brands or types of paints will be permitted.
- c. The VENDOR shall supply Data Sheets for each type of paint which shall include the recommended minimum and maximum overcoating periods for the dry film thickness specified in this Specification.
- d. Paint Systems, i.e. primer, undercoat, finish coat, shall be from the same VENDOR to ensure compatibility.
- e. All Coating Materials used shall conform to the following generalized composition clauses. In meeting the composition clauses, the Paint VENDOR shall provide evidence of compliance and shall produce premium quality paint with the full performance required by this Specification.
- f. Zinc Rich Primers shall be based on two pack epoxy media and shall contain a minimum of 80% metallic zinc by weight in the dry film and comply with SSPC Paint 20 Level 2.
- g. Zinc Phosphate two-pack Epoxy Primer shall be based on polyamide cured two-pack epoxy media and shall contain a minimum of 20% zinc phosphate by weight in the dry film.

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- h. Zinc Silicate Primer shall be based on inorganic or ethyl silicate media and shall contain not less than 80% metallic zinc by weight in the dry film and or complies with the composition and performance requirements of SSPC Paint 20 Level1 and in accordance with ASTM D520 Type II (low lead). The nominated paint System shall pass the accelerated cyclic testing ISO20340.
- i. Two-pack epoxy finishing paints shall be based on polyamide cured tow-pack epoxy media and shall be pigmented with titanium dioxide and/or light fast colored pigments to provide the necessary opacity, film build and weather resistance.
- j. Micaceous iron oxide pigmented paint shall contain a minimum of 25% natural laminar micaceous iron oxide by weight in the dry film.
- k. Aluminium silicone paint shall be based on a one-pack formulation comprising leafing aluminium pigment dispersed in silicone or modified silicone media, the paint shall be capable of withstanding continuous exposure to temperatures between 120°C and 500°C. Only air curing heat resistant aluminium paints shall be applied (Post heat curing Materials are not acceptable).

15.2 Material Storage

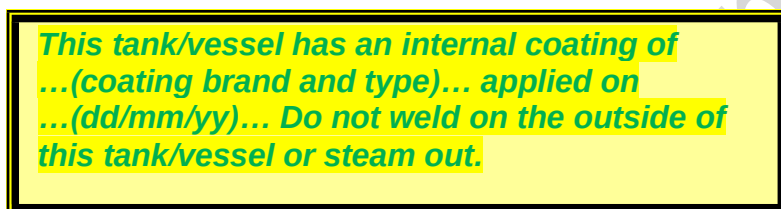
- a. Storage of paint Materials at the site of usage shall be the responsibility of the paint application CONTRACTOR and the storage planning shall take into account the geographical and meteorological conditions and make provisions for any abnormality such as temperatures.
- b. Painting CONTRACTOR shall advise the Painting VENDOR of the intended storage conditions and any recommendations received from the VENDOR shall be included in the storage arrangement.
- c. Records of paint material batch numbers together with dates of receipts of each batch shall be kept by the paint material applicator so that a sequence of storage can be arranged to ensure material is issued from store in the same sequence as received at painting venue. The CONTRACTOR /OWNER/PMC reserves the right to inspect these records and check the issuing organization at any time during the execution of contract.
- d. Storage shall take into account and make provision for governing Local Safety Regulations, adequate fire prevention Equipment and its maintenance and the elimination of ignition risks (e.g. No smoking, use of flame proof Equipment etc).
- e. Purposeful spillage shall be prohibited, but sufficient inert absorbing material shall always be available for the immediate clean-up of accidental spillage. The use of rags and wood dust for clean-up of inflammable Materials is not permitted.
- f. The disposal of waste and empty containers shall be subject to regulations laid down before the commencement of material deliveries. Empty and/or partially empty containers shall be collected and disposed of in accordance with the regulations.
- g. Paint Materials are potentially harmful to operatives and it shall be necessary to maintain a high standard of personnel hygiene. Paint CONTRACTORS shall advise the CONTRACTOR of details of precautions being taken to protect all working personnel in the vicinity of paint storage and handling.

16.0 Numbering and Identification

16.1 Tank, Equipment and Pipe Numbering

- a. Tank and equipment shall be identified by sign writing (not stenciling) externally at two locations with full gloss material compatible with the topcoat applied. Sufficient coats shall be applied to attain full opacity. Unless separate contrast colors are specified in the contract documents all sign writing shall be black in white background.

Where tanks and vessels have an internal coating applied, there shall be a stenciled sign marked on or adjacent to the main manhole stating: Stencil letters shall have enough size to give clearly visible. Letter size to be used is 100 mm high x 15 mm stroke wide painted in black on a contrast background as below example.



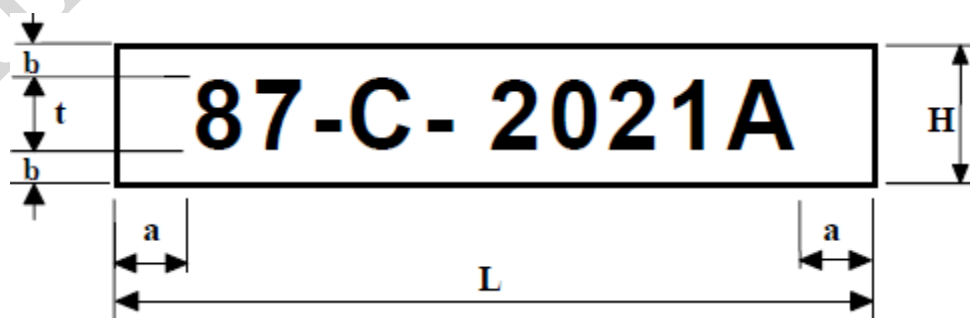
- b. Tanks

Unless otherwise specified in the contract documents and size of numbering shall be as detailed below.

Tank Height (Number of courses)	Location of Lettering	Letter Size (mm) Height X Stroke Width	Orientation
2 & 3	2nd Course	900 X 150	As directed by Superintendent
4	3rd Course	900 X 150	
5 & above	4th Course	1500 X 250	

- c. Equipment

The tag number of each equipment item, such as vessels, heat exchangers, pumps, compressors and furnaces shall be stenciled on the foundation, support skirt or saddle. The letters shall be black on a white background sized as shown below:



Lettering Sizes

H = 300 mm

Basic color = white

Letter color = black

t =200 mm

Stroke width= 30 mm

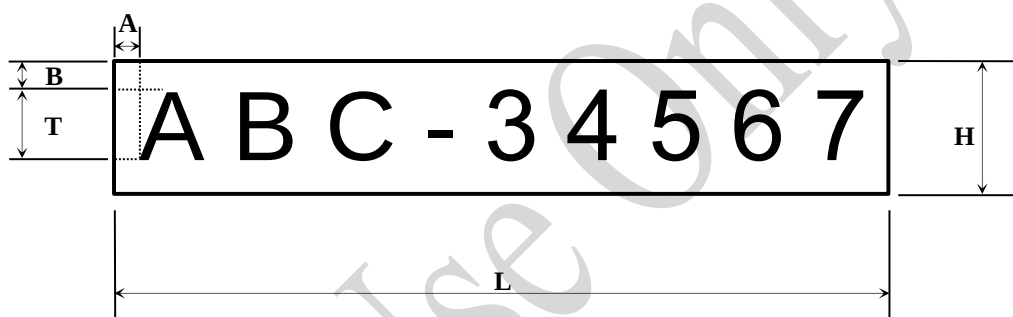
a =50 mm

b =50 mm

L = As required

d. Piping/Line

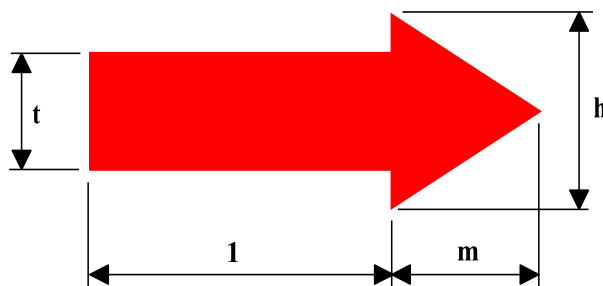
Piping/lines shall be marked with the service and line number at plant battery limits, at every second main pipe bent and at each road crossing. The letters shall be black on a white background and shall be legible from grade sized as shown below. Stencils or tapes may be used. The OWNER representative may nominate the location in some case.



SIZE OF LETTERING FOR PIPING

Pipe (NPS)	L	H	T	Stroke width	A	B
INCH	mm	mm	mm	mm	mm	mm
3/4 - 1	As required	25	15	3	10	5
1 1/2 - 2	As required	30	20	3	10	5
2 1/2 - 6	As required	50	30	5	20	10
8 - 10	As required	90	60	10	30	15
OVER 10	As required	150	100	15	50	25

Black arrows indicating direction of flow shall be located adjacent to valves, flanges, changes in direction and where pipes pass through walls, floors and platforms. Dimensions as below:



For Pipe NPS 4" and below

For Pipe NPS 6" and over

d = pipe diameter

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$$t = d/2$$

$$h = 2t$$

$$m = 2h/3$$

$$l = 2.5m$$

$$t = 9\text{cm}$$

$$h = 15\text{cm}$$

$$m = 10\text{cm}$$

$$l = 20\text{cm}$$

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Table 1.1 For surface preparation by blasting

A) For carbon and low alloy steel

Painting System	Temperature Range (Design Temp.)	Insulation	Primer Coat	Mid Coat	Top Coat
CB-1	-196 °C ~ -1 °C	Uninsulated	Inorganic Copolymer	-	Inorganic Copolymer
CB-2	-196 °C ~ -46 °C	Insulated	Inorganic Copolymer	-	Inorganic Copolymer
CB-3	-45 °C ~ 200 °C	Insulated	Epoxy phenolic / Novolac / Inorganic Copolymer / Alkylated Amine Epoxy	-	Epoxy phenolic / Novolac / Inorganic Copolymer / Alkylated Amine Epoxy
CB-4	0 °C ~ 120 °C	Uninsulated	Inorganic zinc silicate / Zinc rich epoxy primer	Epoxy mid-coat / Polyamide epoxy	Polyurethane
CB-5	121 °C ~ 200 °C	Uninsulated	Inorganic zinc silicate / Zinc rich epoxy primer	Silicone acrylic	Silicone acrylic
CB-6	201 °C ~ 400 °C	Uninsulated	Inorganic zinc silicate / Zinc rich epoxy primer	Aluminium silicone	Aluminium silicone
CB-7	201 °C ~ 400 °C	Insulated	Inorganic Copolymer	-	Inorganic Copolymer
CB-8	Cyclic Services (-50 °C ~ 400 °C)	Both	Inorganic Copolymer	-	Inorganic Copolymer
CB-9	401 °C ~ 650 °C	Both	Inorganic Copolymer	-	Inorganic Copolymer

**B) For stainless steel**

Painting System	Temperature Range (Design Temp.)	Insulation	Primer Coat	Mid Coat	Top Coat
SB-1	-196 °C ~ -1 °C	Uninsulated	Inorganic Copolymer	-	Inorganic Copolymer
SB-2	-196 °C ~ -46 °C	Insulated	Inorganic Copolymer	-	Inorganic Copolymer
SB-3	-45 °C ~ 200 °C	Insulated	Epoxy phenolic / Novolac / Inorganic Copolymer / Alkylated Amine Epoxy	-	Epoxy phenolic / Novolac / Inorganic Copolymer / Alkylated Amine Epoxy
SB-4	0 °C ~ 120 °C	Uninsulated	Epoxy primer	Epoxy	Polyurethane
SB-5	121 °C ~ 200 °C	Uninsulated	Silicon acrylic / Epoxy phenolic / Novolac / Alkylated Amine Epoxy	-	Silicon acrylic / Epoxy phenolic / Novolac / Alkylated Amine Epoxy
SB-6	201 °C ~ 400 °C	Uninsulated	Silicon aluminum / Epoxy phenolic / Novolac	-	Silicon aluminum / Epoxy phenolic / Novolac
SB-7	201 °C ~ 400 °C	Insulated	Inorganic Copolymer	-	Inorganic Copolymer
SB-8	Cyclic Services (-50 °C ~ 400 °C)	Both	Inorganic Copolymer	-	Inorganic Copolymer
SB-9	401 °C ~ 650 °C	Both	Inorganic Copolymer	-	Inorganic Copolymer

Table 1.2 For surface preparation by power tool

A) For carbon and low alloy steel

Painting System	Temperature Range (Design Temp.)	Insulation	Primer Coat	Mid Coat	Top Coat
CP-1	-196 °C ~ -1 °C	Uninsulated	Inorganic Copolymer	-	Inorganic Copolymer
CP-2	-196 °C ~ -46 °C	Insulated	Inorganic Copolymer	-	Inorganic Copolymer
CP-3	-45 °C ~ 200 °C	Insulated	Epoxy phenolic / Novolac / Inorganic Copolymer / Alkylated Amine Epoxy	-	Epoxy phenolic / Novolac / Inorganic Copolymer / Alkylated Amine Epoxy
CP-4	0 °C ~ 120 °C	Uninsulated	Surface tolerant epoxy	Epoxy	Polyurethane
CP-5	121 °C ~ 200 °C	Uninsulated	Silicon Acrylic / Alkylated Amine Epoxy	Silicon Acrylic / Alkylated Amine Epoxy	Silicon Acrylic
CP-6	201 °C ~ 400 °C	Uninsulated	Silicon Aluminum	Silicon Aluminum	Silicon Aluminum
CP-7	201 °C ~ 400 °C	Insulated	Inorganic Copolymer	-	Inorganic Copolymer
CP-8	Cyclic Services (-50 °C ~ 400 °C)	Both	Inorganic Copolymer	-	Inorganic Copolymer
CP-9	401 °C ~ 650 °C	Both	Inorganic Copolymer	-	Inorganic Copolymer

**B) Stainless steel**

Painting System	Temperature Range (Design Temp.)	Insulation	Primer Coat	Mid Coat	Top Coat
SP-1	-196 °C ~ -1 °C	Uninsulated	Inorganic Copolymer	-	Inorganic Copolymer
SP-2	-196 °C ~ -46 °C	Insulated	Inorganic Copolymer	-	Inorganic Copolymer
SP-3	-45 °C ~ 200 °C	Insulated	Epoxy phenolic / Novolac / Inorganic Copolymer / Alkylated Amine Epoxy	-	Epoxy phenolic / Novolac / Inorganic Copolymer / Alkylated Amine Epoxy
SP-4	0 °C ~ 120 °C	Uninsulated	Surface tolerant epoxy	Epoxy	Polyurethane
SP-5	121 °C ~ 200 °C	Uninsulated	Silicon acrylic / Epoxy phenolic / Novolac / Alkylated Amine Epoxy	-	Silicon acrylic / Epoxy phenolic / Novolac / Alkylated Amine Epoxy
SP-6	201 °C ~ 400 °C	Uninsulated	Silicon aluminum / Epoxy phenolic / Novolac	-	Silicon aluminum / Epoxy phenolic / Novolac
SP-7	201 °C ~ 400 °C	Insulated	Inorganic Copolymer	-	Inorganic Copolymer
SP-8	Cyclic Services (-50 °C ~ 400 °C)	Insulated/ Uninsulated	Inorganic Copolymer	-	Inorganic Copolymer
SP-9	401 °C ~ 650 °C	Insulated/ Uninsulated	Inorganic Copolymer	-	Inorganic Copolymer

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Table 1.3 For special application

Painting System	Application	Primer Coat	Mid Coat	Top Coat
Z-1	Carbon steel piping & structural steel locate at marine area under deck, splash zone areas. Z-1 system could applicable for uninsulated piping and equipment where have condensation / wetting surface.	Surface tolerant epoxy	High build epoxy	Polyurethane
Z-2	Galvanized steel.	Surface tolerant epoxy	-	Polyurethane
Z-3	Structural steel under concrete fireproof.	Surface tolerant epoxy	-	Surface tolerant epoxy
Z-4*	General concrete structure surfaces	Epoxy finish	Epoxy finish	Epoxy finish
Z-5*	Anti-slip coating on steel floor	Inorganic zinc silicate / Zinc rich epoxy primer	Solvent free epoxy + Anti-slip material	Polyurethane
Z-6	Fiber reinforced plastic (FRP) & Glass reinforced epoxy (GRE)	-	-	Polyurethane

*When specific requirement

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Table 2: Coating Colors Scheme

Equipment and/or Service	Color Code	Standard RAL nr.
Structural Steel, including platforms and stair beams	Aluminum	RAL 9006
Railing denoting escape routes on process Structures (1 meter either side of opening)	Alert Orange	RAL 2003
Handrails, ladder cages, ladders, Safety Gate.	Zinc Yellow	RAL 1018
Vessels (excl. tanks)	Aluminum	RAL 9006
Tank	White	RAL 9010
Insulated Equipment, Piping and miscellaneous	Aluminium	RAL 9006
Piping (excluding lines listed below)	Aluminum	RAL 9006
Fire water	Flame Red	RAL 3000
Foam	Zinc Yellow	RAL 1018
Instrument Air	Sky Blue	RAL 5015
Potable Water	Leaf Green	RAL 6002
H ₂ S Gas	Zinc Yellow	RAL 1018
Industrial Water	Grey Brown	RAL 8019
Fuel Gas (NG)	Signal Yellow	RAL 1003
Machinery Pumps, electrical Equipment	Window Grey	RAL 7040
Dangerous obstructions	Black and Yellow (in alternative bands)	RAL 9005/1018
Dangerous or exposed parts of Machinery	Pastel Orange	RAL 2003
Fire Equipment and service	Flame Red	RAL 3000
Refractory lined Equipment (temperature up to 400°C)	Aluminium	RAL 9006
Lifting beams	Flame Red	RAL 3000
Overhead travelling Cranes	Zinc Yellow	RAL 1018
Valves	White Aluminium	RAL 9006
Handwheels	Jet Black	RAL 9005

Note:

1. Aluminum color is preferred for insulated and fire proofed objects, no RAL is specified. Where painting is applied in the existing plant, the RAL of existing plant may be used.